



Stage A · Construction (yr 0)

Inputs: raw materials · fossil fuels

A1-A3 Materials & manufacturing

- Gas & oil processing
- Furnaces; AC; heaters

A4-A5 Transport & installation

- Pipeline / truck transport
- Furnaces + oil tanks
- Gas mains; service lines



Outputs: construction waste · emissions

Critical output

fugitive gas leakage
(transmission / distribution)



Stage B · Operation (yr 1-50)

Inputs: spare parts · electricity · gas & oil

B2-B4 Maintenance & replacement

- Filters; burner; tank checks
- Furnace/AC swap (20 & 40 yr)

B6 Operational energy

- 70/30 gas : oil combustion
- Electricity: AC + fans
- Heating, cooling & DHW



Outputs: combustion CO₂, CH₄, N₂O

Critical output

behind-the-meter + 2.04%
distribution CH₄ leakage



Stage C · End of life (yr 50)

Inputs: energy for decommissioning

C1-C2 Decommissioning & removal

- Furnace/AC removal
- Units & tanks to scrap

C3-C4 Disposal

- Landfill components
- Gas-line left in place
- Oil-tank disposal



Outputs: final waste · soil-contamination risk

Note

net end-of-life ≈ 0
(abandonment in place)